

Financial Analysis and Decision Making

Global BBA Fall 2026

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Session 4: Rates

Housekeeping

About me

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Excel Functions

- ▷ `-PV(rate,nper,pmt)` & `-FV(rate,nper,pmt)`
 - Gives the PV or FV of an annuity
- ▷ `=PMT(rate,nper,pv)`
 - Solves for C in the annuity formula
- ▷ `=NPV(rate,value1,[value2],...)`
 - Assumes $[value1]$ is C_1

Excel Functions

	A	B	C	D	E	F	G
1	C_0	C_1	C_2	C_3	C_4	C_5	C_6
2	-100	50	50	50	50	50	50
3	Discount Rate	10%					
4	NPV	83					
5	IRR	45%					

Formula in cell B4: = A2 + NPV (B3, B2 : G2)

Formula in cell B5: = IRR (A2 : G2)

Excel Functions

	A	B	C	D	E	F	G
1	C_0	C_1	C_2	C_3	C_4	C_5	C_6
2	-100	50	50	50	50	50	50
3	Discount Rate	10%					
4	NPV	83					
5	IRR	45%					

Alternatively in cell B4: = A2 - PV(B3, COLUMNS(B2:G2), B2)

where PV(rate,nper,pmt)

Excel Functions

	A	B	C	D	E	F	G
1	C_0	C_1	C_2	C_3	C_4	C_5	C_6
2	-100	?	?	?	?	?	?
3	Discount Rate	10%					
4	NPV	83					
5	IRR	45%					

What if we want to know C (assuming $C_1, \dots, C_6$ are equal)?

= - PMT (B3, COLUMNS (B2 : G2) , B4-A2)

Example

Suppose the present value of an apartment you would like to purchase is \$500,000. You plan a 30-year mortgage with equal annual payments, at an interest rate of 5% compounded annually. What is your yearly payment?



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Example

Suppose the present value of an apartment you would like to purchase is \$500,000. You plan a 30-year mortgage with equal annual payments, at an interest rate of 5% compounded annually. What is your yearly payment?

$$PV = \frac{C}{r} \left[1 - \left(\frac{1}{1+r} \right)^T \right]$$

$$PV \times r = C \times \left[1 - \left(\frac{1}{1+r} \right)^T \right]$$

$$\frac{PV \times r}{\left[1 - \left(\frac{1}{1+r} \right)^T \right]} = C$$

Example

Suppose the present value of an apartment you would like to purchase is \$500,000. You plan a 30-year mortgage with equal annual payments, at an interest rate of 5% compounded annually. What is your yearly payment?

$$C = \frac{PV \times r}{1 - (1 + r)^{-T}}$$
$$C = \frac{500 \times 0.05}{1 - (1.05)^{-30}} = 32.526$$



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Effective Annual Rate

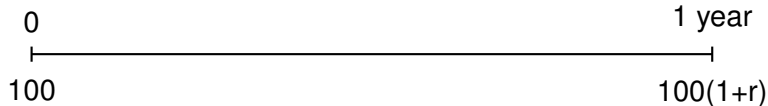
Suppose a bank offers you .001% interest compounded every 6 hours. Is this a lot? A little? As a means of standardization, the **effective annual rate (EAR)** gives you the amount of interest that will be earned at the end of one year.



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Annual Compounding

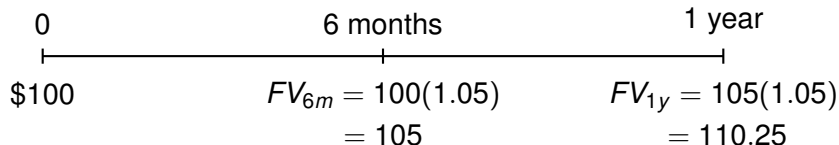


So far, whenever we solved problems using a given r , we assumed *annual compounding*.

Example. How much money would you have in your bank account if you deposit €100 in your bank account at an interest rate of 10%?

Semi-Annual Compounding

What if your bank offered you an annual interest rate of 10%, compounded semi-annually?



You made **10.25%** after 1 year, NOT 10%.

We call 10.25% the **effective annual rate**.

Quoting Rates

“Suppose a bank offers you a semi-annual rate of 10%”

≠

“Suppose a bank offers you an annual rate of 10%, compounded semi-annually”

≠

“Suppose a bank offers you an effective annual rate of 10%”

Quoting Rates

In general, the standard way of quoting rates is

$r\%$ annually, compounded m times a year

such that you earn

$$\text{Deposit} \times \left(1 + \frac{r}{m}\right)^m$$

where compounded

- ▷ daily: $m = 365$
- ▷ monthly: $m = 12$
- ▷ quarterly: $m = 4$
- ▷ annually: $m = 1$

Quoting Rates

In general, the standard way of quoting rates is

$r\%$ annually, compounded m times a year

such that you earn

$$\text{Deposit} \times \left(1 + \frac{r}{m}\right)^m$$

and

- ▷ r : the nominal rate, or the annual percentage rate (APR)
- ▷ $\frac{r}{m}$: the proportional rate

Effective Annual Rate

We want

$$\text{Deposit} \times \left(1 + \frac{r}{m}\right)^m = \text{Deposit} \times (1 + \text{EAR})$$

Exercise 5.4

Compute the EAR for each investment choice

- a. 9% APR compounded monthly (9.38%)
- b. 12% APR compounded annually (12%)
- c. 9% APR compounded daily (9.416%)

Exercise 5.6

Your bank account pays interest with an EAR of 9%. What is the APR quote for this account based on semiannual compounding? What is the APR with monthly compounding?

(8.806%,8.649%)



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Equivalent Discount Rate

The effective annual rate, as the name implies, is the total amount of interest you receive after one year.

Define the **equivalent n -period discount rate** as the total amount of interest you receive after time n .

* For all time-value of money calculations, use the EAR or the n -period discount rate.

Equivalent Discount Rate

The EAR and the n -period equivalent discount rate is related in the following way

$$(1 + \text{EAR})^n = 1 + r_{\{n\}}$$

where $r_{\{n\}}$ denotes the n -period equivalent discount rate.

Example. Suppose the EAR is 10%. What is the equivalent 6-month discount rate? (2.47%)

Equivalent Discount Rate

Example. Suppose the equivalent 2-year discount rate 15%. What is the effective annual rate?

(7.238%)



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Strategy for Solving Questions

1. Convert the interest rate to an effective annual rate
2. Find the appropriate n -period discount rate using

$$(1 + \text{EAR})^n = 1 + r_{\{n\}}$$

- If necessary, use the relevant n -period discount rate to discount cash flows that arrive every n -periods



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Exercise 5.1

Your bank is offering you an account that will pay 10% interest in total for a two-year deposit. Determine the equivalent discount rate for a period length of:

- a. Six months

(EAR = 4.88% and equivalent 6-month discount rate = 2.41%)



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Exercise 5.1

Your bank is offering you an account that will pay 10% interest in total for a two-year deposit. Determine the equivalent discount rate for a period length of:

- b. One year
- c. One month
(0.397%)



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Exercise

Suppose the equivalent 6-month discount rate is 4%. What is the equivalent 2-year discount rate?

(EAR = 8.16% and equivalent 2-year discount rate = 16.985%)



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Exercise

Suppose a bank offers you an APR of 10% compounded semi-annually. What is the equivalent 1-month discount rate?
(EAR = 10.25% and equivalent 1-month discount rate = 0.816%)



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Exercise

You can buy the car for \$65,000 up front, or you can lease it for \$1,800 per month for 36 months. The firm can borrow at an interest rate of 8% APR with quarterly compounding. Should you purchase the system outright or pay \$1,800 per month?

(equivalent 1-month discount rate = 0.006627% and $PV=57,486$)



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Exercise 5.2

Which do you prefer: a bank account that pays 8% per year (EAR) for three years or:

- a. An account that pays 4% every six months for three years?
(.2653 vs .2597)
- b. An account that pays 0.8% per month for three years?
(.3322 vs .2597)



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Plan For Today

- ▷ Rates
- ▷ Bond Terminology



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Bonds

A bond is a **debt security** in which the issuer/seller (borrower) raises money from investors today in exchange for a promised repayment at the maturity date.

Example. In July 2026, Amazon raised \$25bn through a bond sale to finance its capital expenditures in AI investments. Terms include:

- ▷ Maturities ranging from 3 to 40 years
- ▷ At 65 to 145 basis points (.65-1.45%) above U.S. treasuries
- ↔ Coupon rates between 4.6 – 6.25%



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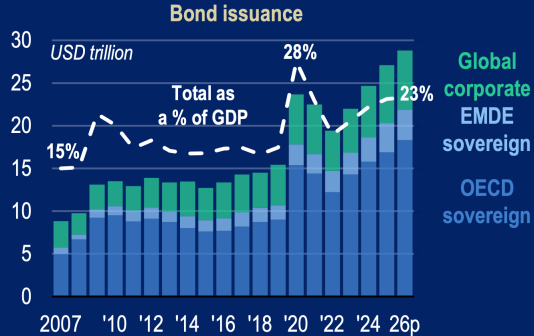
Some Facts: Corporate vs Govt Bonds



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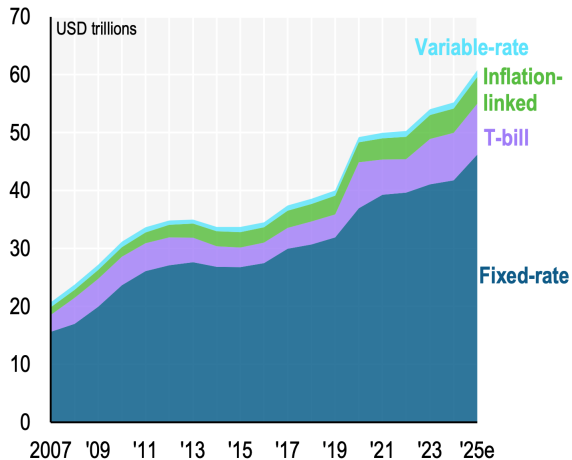
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Sovereign and corporate bond issuance reached a record in 2025, with USD 29 trillion projected for 2026



Source: Global Debt Report 2026, *OECD*

Some Facts: Total Issuance of Govt Bonds



Source: Global Debt Report 2026, *OECD*



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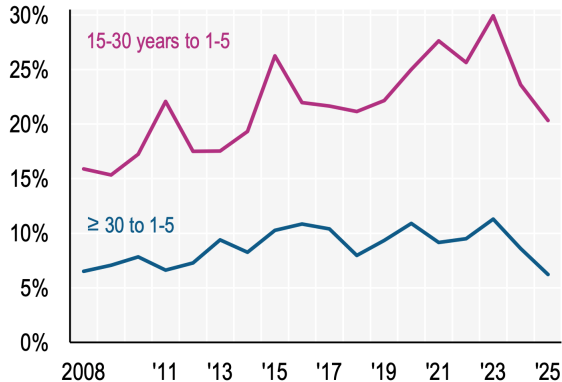
Some Facts: Maturity of Govt Bonds



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**A. Ratio of long-term to short-term bond
issuances in the OECD area**

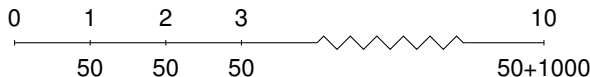


Source: Global Debt Report 2026, *OECD*

Bond Terminology

Bonds typically pay periodic **coupon payments** as a source of intermediate cash flows for investors. At maturity, the issuer repays the principal, or the **face value** of the bond.

Example. Suppose the U.S. issues a 10-year treasury note with a face value of \$1000 and a coupon rate of 5%. Then, investors will receive



▷ !! The coupon rate $\neq$ the interest rate on the bond.

Corporations. Investment-grade, junk, callable, convertible bonds.

Governments. Treasury bills, notes, bonds, municipal bonds (local)

Zero-Coupon Bond

A **zero-coupon** bond is a bond that only repays the face value at maturity (there are no coupon payments).

Example. Suppose the U.S. issues a 5-year a zero-coupon bond with face value of \$1000. If investors are willing to pay \$783.53 for this bond, what is the implied rate of interest?

$$\begin{aligned}\text{price} &= \frac{\text{face value}}{(1 + \text{rate})^t} \\ 783.53 &= \frac{1000}{(1 + r)^5} \\ r &= \left(\frac{1000}{783.53} \right)^{1/5} - 1 = 5\%\end{aligned}$$

How to interpret? An investor pays \$783.53 today to receive \$1000 in 5 years. This gives a return of $(1000 - 783.53) / 783.53 = 27.6\%$, which, on an annual basis, is 5%.

Spot rate

The **spot rate** is the discount rate that equates the present value of a zero-coupon bond to its current market price.

The spot rate computed for a ZCB maturing in N years is referred to as the N -year spot rate. We can alternatively think of the spot rate as the equivalent N -period discount rate.

* From the previous example, we would say that the 5-year spot rate is 5%.



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Example

What is the price of a 2-year zero-coupon bond with a face value of \$1,000? Assume the 2-year spot rate $\Leftrightarrow$ the equivalent 2-year discount rate is 5%.

($p = 907.03$)



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Coupon Bond

A **coupon-bearing** bond is a bond that pays the holder periodic interest payments (called coupons), paid typically semi-annually or annually. The value of a coupon payment is given by:

$$\text{Coupon} = \frac{\text{Coupon Rate} \times \text{Face Value}}{\text{Number of payments per year}}$$

- ▷ The coupon rate is always quoted as an **annual rate**



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Example

Suppose France issues a 5-year coupon bond with face value of €1000 and a coupon rate of 6% paid semi-annually.

- a. What cash flows will you receive?



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Example

Suppose France issues a 5-year coupon bond with face value of €1000 and a coupon rate of 6% paid semi-annually.

- a. Assume the effective annual rate is 10%. What is the value of this bond?



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Example

Suppose France issues a 5-year coupon bond with face value of €1000 and a coupon rate of 6% paid semi-annually.

- a. Assume the effective annual rate is 10%. What is the value of this bond?

If the EAR is 10%, then the equivalent 6-month discount rate is

$$\begin{aligned}(1 + \text{EAR})^n &= 1 + r_{\{n\}} \\ (1.1)^{0.5} &= 1 + r_{\{6m\}} \\ r_{\{6m\}} &= 4.88\%\end{aligned}$$

With semi-annual coupon payments, there are 10 payments over 5 years, plus the face value of the bond



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Example

Suppose France issues a 5-year coupon bond with face value of €1000 and a coupon rate of 6% paid semi-annually.

- a. Assume the effective annual rate is 10%. What is the value of this bond?

With semi-annual coupon payments, there are 10 payments over 5 years, plus the face value of the bond. Using the formula for annuities, we have

$$p = \underbrace{\frac{C}{r} \left[1 - \left(\frac{1}{1+r} \right)^T \right]}_{\text{PV of coupon payments}} + \underbrace{\frac{F}{(1+r)^T}}_{\text{PV of face value}}$$

$$p = \frac{30}{.0488} \left[1 - \left(\frac{1}{1.0488} \right)^{10} \right] + \frac{1000}{(1.0488)^{10}}$$

$$p = 853.98$$



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Exercise

Consider a 30-year coupon-bond with a 5% coupon rate paid annually and a face value of \$100. Assume a 5% APR with semi-annual compounding. What is the present value of this bond?

$$p = 99.04$$



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